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Visual Correspondence Hallucination[1]

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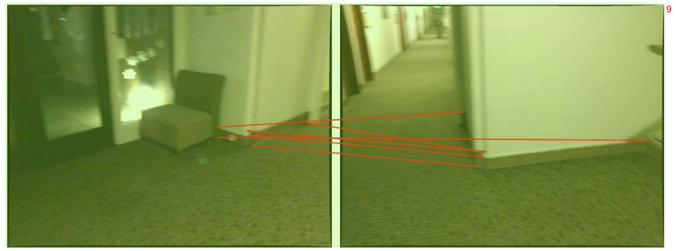
ICLR

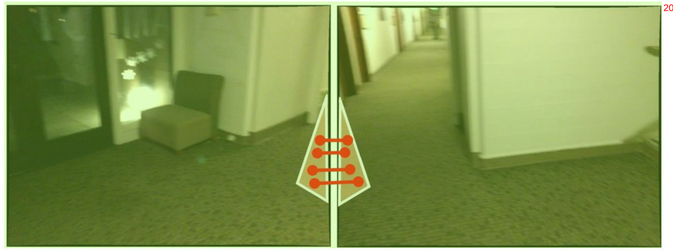
1. Low-overlap keypoint matching[3]

- Keypoint matching is key for fundamental matrix & camera pose estimation[5]
- In low-overlap scenarios however, finding correspondences is very challenging[6]
- Existing methods will tend to find correspondences in non-covisible areas[7]

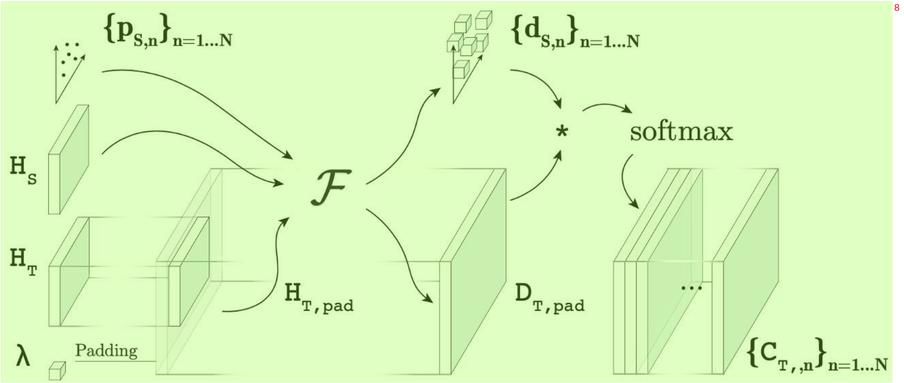


SuperPoint + SuperGlue[10]

- Correct matches within image boundaries provide poor constraints[15]



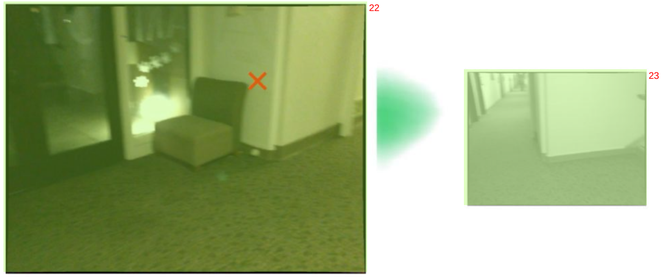
2. Introducing NeurHal[4]


[8]

- We predict correspondence maps on extended field-of-views[11][12]
- Training is done on both covisible and non-covisible samples[13]
- We resort to a Transformer-based image processing backbone[14]

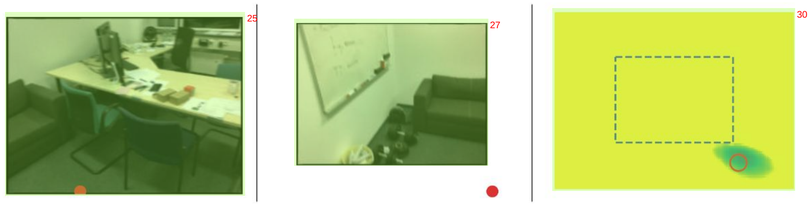
3. Application to absolute camera pose estimation[16]

- We provide quantitative analysis of NeurHal's ability to hallucinate[17]



NeurHal on validation image pair (novel scene)[29]

- NeurHal achieves sota on absolute pose estimation for low-overlap image pairs[18][19]
- More results in our paper ![21][24]



Source[26] Target[28] Predicted NRE Map[31]